Soumyajyoti Dutta

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Machine Learning for Cybersecurity | Small and Large Language Models | Automated Program Synthesis

Education

Texas A&M University — College Station, TX
Ph.D. in Computer Science, Department of CSCE — *Spring 2024 – Fall 2029*
- **Advisor:** Dr. Marcus Botacin
- **Thesis:** LLMs for Automated Rule Generation and Model Interpretability in Cyberdefense.

Texas A&M University — College Station, TX
M.S. in Computer Engineering, Department of ECE — *Fall 2022 – Fall 2023*
- **Coursework:** ML Theory, Deep Learning, Data Mining, Randomized Algorithms, ML-Based Cyberdefenses.

SRM University — Chennai, India
B.Tech. in Electronics and Communication Engineering — *Fall 2017 – Spring 2021*

Expertise

Core Competencies:

- End-to-end ML pipelines: large-scale datasets, feature engineering, training, evaluation, deployment.
- Classical ML: classification, clustering, regression, anomaly detection, forecasting.
- Distributed training: trained 50+ models on multi-node GPU clusters (DDP/NCCL optimization).
- Data analysis: statistical analysis and visualization for actionable insights.
- Systems optimization: cache design, tokenization efficiency, memory-optimized loaders.

Technical Stack:

Python (NumPy, Pandas, scikit-learn, Matplotlib, PyTorch, TensorFlow, HuggingFace), C++, Rust, YARA, Django/Flask, Git, AWS, SLURM

Professional Experience

Texas A&M University — College Station, TX
Research Assistant (Affiliation: Botacin's Lab) — *Spring 2024 – Present*
- Designed curriculum and trained model families spanning 140M–9B parameters (BART, T5, Nemotron 3 Nano 9B, Gemma 3 270M/1B/4B, LLaMA 3.1 8B, LLaMA 3.2 1B/3B) for structured code generation using DeepSpeed, FSDP, and CUDA on multi-node SLURM GPU clusters.
- **YaraGen** (USENIX 2026 submission in preparation):
  - Built end-to-end LLM pipeline for YARA rule generation using curriculum learning, structural reinforcement and AST-embedded tokenization.
  - Developed YARA syntax-aware tokenizer; reduced token-space by 10×.
  - Introduced curriculum-based difficulty scheduling using rule types and complexity tiers.
  - Designed *Brownie & Puff* Evaluation Function (BLEU + Parsing + SAT semantics) for training-time inference evaluation; benchmarked across T5, BART, and LLaMA model families.
  - Built BiLSTM and Transformer Encoder NER pipeline for IOC extraction from unstructured PDF threat reports.
  - Developed scalable dataset framework (translation, patching, IOC detection) producing 100M+ labeled training examples.
- **AutoPYara** (open-sourced, ACSAC 2026 submission):
  - Re-implemented biclustering-based rule generation; evaluated multiple clustering algorithms.
  - Achieved 14% performance gain via optimized centroid selection, 10% via sub-clustering, and 8% improved generalization in low-sample threat-hunting scenarios.

Texas A&M University — College Station, TX
Graduate Student Researcher — *Summer 2023*
- Mathematical modeling of prostate cancer genomic pathways; simulated multi-drug interactions.
- Supported biologists through computational simulations for model validation.

Cognizant Technology Solutions — Kolkata, India
Junior Software Engineer — *2021 – 2022*
- Developed scalable full-stack web applications for logistics automation (Client: TJX Companies).
- Optimized REST endpoints for latency & concurrency and collaborated with QA/DevOps on CI/CD pipelines.

Projects

Machine Learning Malware Detection Model | *1st Place*

- End-to-end ML system for Windows PE classification.
- Integrated EMBER, BODMAS, Benign-NET datasets (2M+ samples).
- Extracted PE-header and byte-level features using LightGBM/XGBoost.
- Achieved 96.2% accuracy, 0.97 F1; designed best-performing adversarial attacks.

Multiperspective Hawkeye

- Implemented ISCA'16 cache replacement policy in zsim.
- Extended with PC-tracking and multi-policy benchmarking.

HelloPentagon

- Explainable ML-based malware defense chatbot integrating ML classification with LLM triage.

Carotid Artery USG Analysis

- CNN-based ultrasound classification (Springer Chapter, 2022).

WildFire

- Spatiotemporal ML modeling for wildfire spread prediction using climate and satellite data.

Fashion-MNIST

- CNN classifier comparing ResNet/VGG architectures with regularization analysis.

Teaching & Outreach

Teaching Assistant (CSCE 439/704) — Texas A&M University
Data Analytics for Cybersecurity — *Spring 2026*

- Core data analytics foundations: clustering, supervised ML, anomaly detection, and security-focused data visualization applied to attacks, vulnerabilities, and adversarial behaviors.
- Supported lectures & grading, developed assignments, and provided project guidance

Partner & Presenter — Texas A&M University
TAMU CS Day — *2024, 2025*

- Conducted interactive demos on cybersecurity and AI research for high school students.

Teaching Assistant (ECEN 325) — Texas A&M University
Electronics — *Summer 2023*

- Supported lectures, grading, and lab sessions.

Technical Coordinator — Texas A&M University
TAMU Student Research Week — *Spring 2023*

- Automated judging and submission infrastructure.